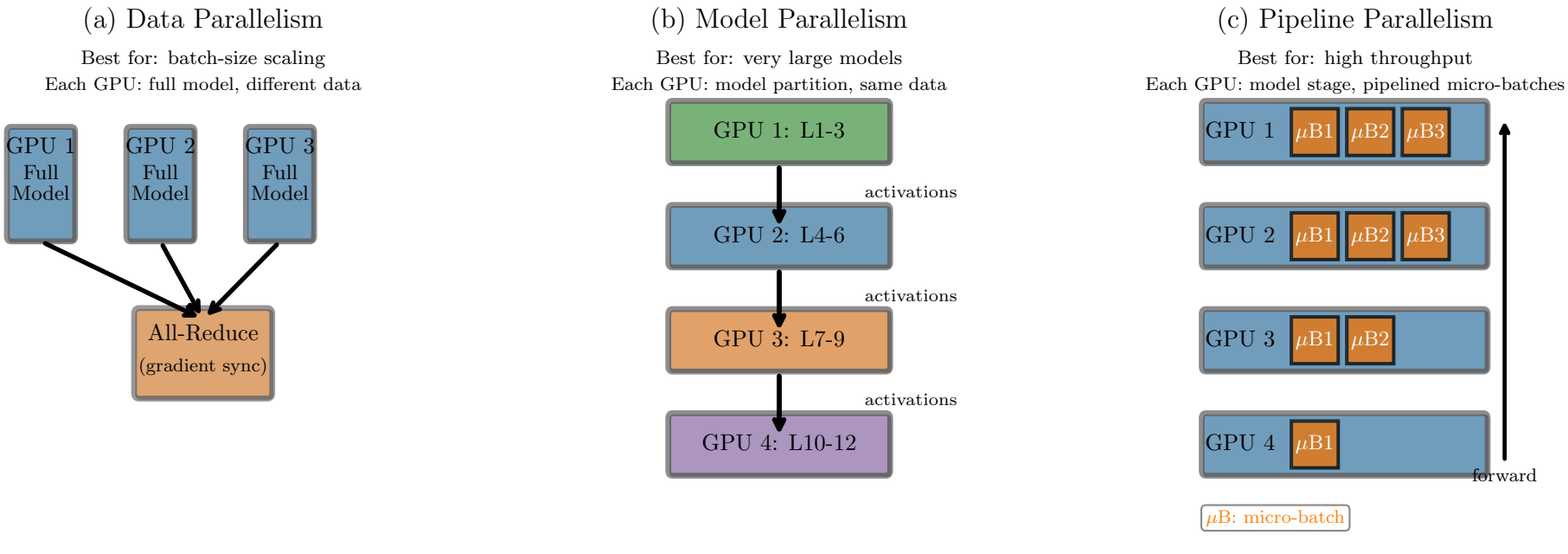
(a) Data Parallelism
Best for: batch-size scaling
Each GPU: full model, different data
GPU 1 Full Model
GPU 2 Full Model
GPU 3 Full Model
All-Reduce (gradient sync)
(b) Model Parallelism
Best for: very large models
Each GPU: model partition, same data
GPU 1: L1-3
activations
GPU 2: L4-6
activations
GPU 3: L7-9
activations
GPU 4: L10-12
(c) Pipeline Parallelism
Best for: high throughput
Each GPU: model stage, pipelined micro-batches
GPU 1 μB1 μB2 μB3
GPU 2 μB1 μB2 μB3
GPU 3 μB1 μB2
GPU 4 μB1
forward
μB: micro-batch